1. Draw the worldline of an object that remains at rest at a distance a a from the origin of a light cone in a spacetime diagram. [2]

2. Why don't atoms, humans, or galaxies expand along with the universe? Where does the argument based on the Friedmann equation break down? [1+1]

3. If the universe is expanding, what is it expanding into? Explain the meaning of the expansion. Has the universe an edge? [1+1+1]

4. Explain the difference between comoving distance and physical distance. Which one changes with time? [1+1]

5. A stick of one meter is at rest in a frame O'. It makes an angle 30° with respect to the X' axis. There is another frame O that moves relative to O' along the $X - X'$ axis ($X \| X'$, $Y \| Y'$ and $Z \| Z'$). What must be the relative velocity v of the stick if it makes an angle 45° with respect to the X-axis of O? [7]

6. A train 1Km long, as measured by an observer on the train, travels at a speed of $0.95c$, where c is the light speed. Two lightning bolts strike the ends of the train simultaneously as determined by an observer on the ground. What is the time separation by an observer on the train? [7]

7. Prove that if if a light ray is emitted from the surface of earth and received at a height h then wavelength is redshifted. Calculate the shift in wavelength. [7]